

CS515 Coding Project Report

Team Members:

- Ahmed Gameil
- Mohammed Mostafa

Selected Algorithm

- Caesar Cipher, Rail Fence Cipher, Row Transposition Cipher

Project Description

- This project is a command-line tool developed in C++ that demonstrates classical encryption and decryption techniques. It allows users to input plaintext and a key to produce ciphertext, and vice versa.

Algorithms Explanation

1. Caesar Cipher

- A substitution cipher where each letter is shifted by a fixed number (key).

2. Rail Fence Cipher

- A transposition cipher that writes text in a zigzag pattern across multiple rows.

3. Row Transposition Cipher

- Text is written in rows and rearranged based on a key.

Program Features

- Supports encryption and decryption
- Handles uppercase and lowercase letters
- Command-line interface
- Multiple cipher options

How to Run

A pre-compiled version of the cipher commands and the shell are available in the output folder.

You can use bash script "build.sh" to recompile the commands and shell.

Shell is a user-friendly CLI that collects arguments from the user and pass them to the sub-commands.

Besides the shell, each cipher has its own command that can be called directly with the required parameters. Use <command> --help to show available options.

Screenshots

```
+-----+
| Session Summary |
+-----+
| Cipher | Caesar (csr) |
| Action | encrypt     |
| Input type | direct    |
| Input   | Hello World From Cipher Pro... |
| Key     | 5         |
| Case mode | capital   |
| Remove spaces | yes      |
| Output   | stdout (console) |
+-----+
| Generated command |
| C:\output\windows\64-bit\csr.exe "Hello World From Cipher P |
| roject" --action encrypt --key 5 --mode capital --remove-sp |
+-----+
| * Spaces will be stripped from output before processing. |
| Running cipher... |
+-----+
| MJQQTBTWQIKWTRHNUMJWUNTOJHY |
+-----+
```

```
+-----+
| Session Summary |
+-----+
| Cipher | Caesar (csr) |
| Action | decrypt     |
| Input type | direct    |
| Input   | MJQQTBTWQIKWTRHNUMJWUNTOJHY |
| Key     | 5         |
| Case mode | capital   |
| Remove spaces | yes      |
| Output   | stdout (console) |
+-----+
| Generated command |
| C:\output\windows\64-bit\csr.exe MJQQTBTWQIKWTRHNUMJWUNTOJH |
| Y --action decrypt --key 5 --mode capital --remove-spaces |
+-----+
| * Spaces will be stripped from output before processing. |
| Running cipher... |
+-----+
| HELLOWORLDFROMCIPHERPROJECT |
+-----+
```

All screenshots are available on github: https://github.com/maamostafa/cs515_ciphers/tree/main/screenshots

Conclusion

This project demonstrates classical encryption techniques and highlights the differences between substitution and transposition ciphers through practical implementation using C++.